PROJECT SUMMARY

Analyzing E-Learning Platform Purchases using MySQL

Project Overview

This project uses MySQL to analyze sample purchase data from an online learning platform. The database contains three related tables—**learners**, **courses**, and **purchases**—to connect learner details with course information and transaction records. The goal is to understand learner spending, course demand, category revenue, and purchasing behaviour.

Objectives

- Design a relational database with primary-key and foreign-key relationships.
- Insert and validate sample learner, course, and purchase records.
- Use INNER JOIN, LEFT JOIN, and RIGHT JOIN to combine related data.
- Apply aggregation, filtering, sorting, and calculations to answer business questions.

Tools and Techniques

Tool: MySQL / MySQL Workbench **SQL concepts:** CREATE DATABASE, CREATE TABLE, INSERT, primary and foreign keys, JOINs, SUM(), COUNT(DISTINCT), GROUP BY, HAVING, ORDER BY, LIMIT, COALESCE, and FORMAT().

Key Findings from the Sample Data

Metric	Result
Total sample purchase revenue	INR 25,000.00
Highest-revenue category	Data Analytics — INR 13,000.00
Category revenue	Programming — INR 7,500.00; Database — INR 4,500.00
Top course quantities	SQL for Beginners, Python Programming, and Power BI Dashboard — 3 each
Learners purchasing across multiple categories	Arun Kumar, Priya Sharma, and John Smith
Course with no purchases	Java Programming

Recommendations

Prioritize promotion of Data Analytics courses because this category generated the highest sample revenue. Investigate why Java Programming received no purchases and consider improving its visibility, pricing, or course offering. Use learner purchase patterns to recommend courses from complementary categories and encourage repeat purchases.

Conclusion

The project demonstrates how a structured MySQL database and analytical SQL queries can turn transaction records into useful business insights. The findings above are calculated from the illustrative sample records created for this assignment; they should not be interpreted as actual platform performance.